Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Continuous Assessment Test (CAT) – II - October 2025

rogramme	:	B.Tech. (CSE and its Specialization)	Semester	:	Fall semester 2025-2026
:ourse Code :ourse Title	:	BCSE355L , Cloud Architect Design	Slot	:	B1+TB1
'aculty	:	Dr. PRADEEP K Dr. S VIGNESHWARI Dr. TAMILARASI K Dr. SELLAM V Dr. POORNIMA S Dr. DEEPA NIVETHIKA Dr. RAJESH KUMAR Dr. SUGUNA Dr. LINDA JOSEPH	Class Number	:	CH2025260102018 CH2025260101350 CH2025260102012 CH2025260102007 CH2025260102048 CH2025260101473 CH2025260101673 CH2025260101670 CH2025260101668
)uration	:	90 Minutes	Max. Mark		50

Jeneral Instructions: < Use this space to provide additional information such as graph sheet, data book etc.>

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1	a	A financial technology company is building a new application that requires a highly available, scalable, and durable database solution. The application will handle millions of read requests daily, with a high volume of transactions that must be written to the database in case of an infrastructure failure. They are considering using PostgreSQL database. Given these requirements, Design a database architecture that incorporates both PostgreSQL database Multi-AZ Deployments and PostgreSQL database Read Replicas. Explain the role of each component, describe the potential data consistency challenges and how to manage them. (5 Marks)	10	CO4	K3
	b	Detail how this architecture provides read and write scalability, and walk through the step-by-step process of an automatic failover? (5 Marks)			
2	a	A multinational online gaming platform must store extensive player profiles and log millions of real-time gaming sessions across North America, Europe, and Asia-Pacific. During rapid user growth, the company seeks a cloud database solution that can instantly sync changes to profile data and session logs across all regional applications, ensuring low-latency access even during peak traffic and providing continuous availability despite potential outages in single geographic locations. Which type of database would you recommend to address the platform's	10	CO1	K3

	b	requirement for real-time synchronization, high availability, and seamless scalability across multiple regions? Justify your choice with reference to core architectural strengths and trade-offs such as eventual consistency, flexibility, and performance under distributed access patterns. (6 Marks) How should the chosen database solution handle potential conflicts when concurrent updates to a player's achievement record originate from two different regions? What impact does this have on user experience and data consistency? (4 marks)			
3	a b	A fintech startup has been running its workloads on AWS for 9 months. The finance team reports that costs are rising unpredictably, sometimes spiking by 40% compared to projections. A preliminary check shows the following: i. The team uses on-demand EC2 instances for both production and testing. ii. Some applications generate large volumes of logs, which are all stored in S3 Standard without lifecycle policies. iii. The company has no system to track costs by department, making it hard to hold teams accountable. iv. When asked, developers say they are "not sure how billing really works."? Why unpredictable costs are a common challenge in cloud environments compared to on-premise infrastructure? Explain two practices that are contributing to the rising costs. (4 Marks) List AWS Cost Management tools would best help the company uncover hidden spending patterns, and why? Propose a cost governance plan that balances developer freedom with financial accountability. (6 Marks)	10	CO3	K5
4	a b	A healthcare provider is migrating its electronic health record (EHR) system to AWS (S3, RDS, AWS Lambda). The provider must comply with HIPAA and internal audit requirements, and is evaluating key management strategies using AWS KMS. The provider has to choose between AWS-managed, customer-managed, and external key stores. Explain the differences and recommend which key store type is most appropriate for compliance-heavy workloads requiring full control and auditability?. Describe the role of data encryption keys in AWS KMS and how they interact with customer master keys (CMKs)? (5 Marks) Apply your understanding to list AWS services that benefit from automatic key rotation using AWS-managed KMS keys. State the frequency at which AWS-managed KMS keys are automatically rotated for built-in services. (5 Marks)	10	CO2	K5
5	a b	A logistics firm with global operations wants to integrate IoT devices that send delivery status updates every few seconds. Propose a complete workflow that uses AWS service to handle scalable numbers of messages reliably, and processing by a subsequent synthesis process by AWS services. Draw a diagram and explain the AWS service that can be used to access large number of message by the synthesis process. (6 Marks) If the synthesis process could not process few of the delivery message maximum allowed processing times (e.g. 4 times), explain the way the AWS service will handle those messages. (4 Marks)	10	CO1	K6

**********All the best **********